

## **DCA1000EVM Data Capture Card**

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## Trademarks

MicroSD is a trademark of SD-3C, LLC.

## 1 Introduction

The DCA1000EVM is a capture card for interfacing with Texas Instrument's 77GHz xWR1xxx EVM that enables users to stream the ADC data over Ethernet. This design is based on Lattice FPGA LFE5UM-85F-8BG381I with DDR3L. The signal interface between the capture card and the xWR1xxx EVM uses the 60-pin Samtec high density connector.

The following use cases can be accessed using the DCA1000EVM with TI's xWR1xxx EVM.

- LVDS over Ethernet streaming:
  - Raw mode: In this mode, all LVDS data is captured as it is and streamed over the Ethernet interface.
  - Data separated mode: In this mode, the user can add specific headers to different data types; FPGA separates out different data types based on the header and streams it over Ethernet interface.

### 1.1 Key Features

The key features of the DCA1000EVM are:

- The DCA1000EVM is based on LATTICE's High-Performance FPGA LFE5UM-85F-8BG381I
  - Package: 381 pins CABGA (17×17 mm).
  - 84K LUT's, 3744 Kbits Block RAM and 4 PLLs
- 2GByte of DDR3L device
- 16MByte SPI Flash for storing FPGA images
- MicroSD™ Card interface (presently not supported)
- JTAG interface through FTDI FT2232HL Chip
- USB interface to provide SPI, UART, and I2C interfaces through the onboard FTDI device. This is used for mmWave Studio.
- 9 user LEDs, 4 push-button switches, and 1×8 and 1×3 DIP switches for user configurations
- 1Gb/s RGMII Ethernet interface with RJ45 jack
- TI's xWR1xxx EVM provides the below mentioned interfaces on the 60-pin SAMTEC connector QTH-030--01-L-D-A. The connector QTH-030-01-L-D-A can be mated with TI's xWR1xxx EVM using board-to-board cable assembly (Samtec coax micro ribbon cable - Part Number# HQCD-030-02.00-SEU-TBR-1, is included in the kit).
  - SPI interface
  - UART interface
  - I2C interface
  - 4-Lane LVDS interface
  - GPIO signals
  - DMM interface
- The DCA1000EVM is powered up using 5-V power, either from an external DC adapter, or from TI's xWR1xxx EVM, by selecting the power selection switch (SW3).
- DC-DC regulators and LDO regulators are used to derive the required power required for FPGA, DDR3L, and other onboard peripherals.
- RoHS, Pb-free, and REACH-compliant design
- LVDS data rates support a maximum of 600 Mbps. It supports both 4 lanes (xWR1243/1443) and 2 lanes (xWR1642)
- Temperature range – industrial grade
- Maximum current rating – 700-mA maximum

## 1.2 Kit Contents

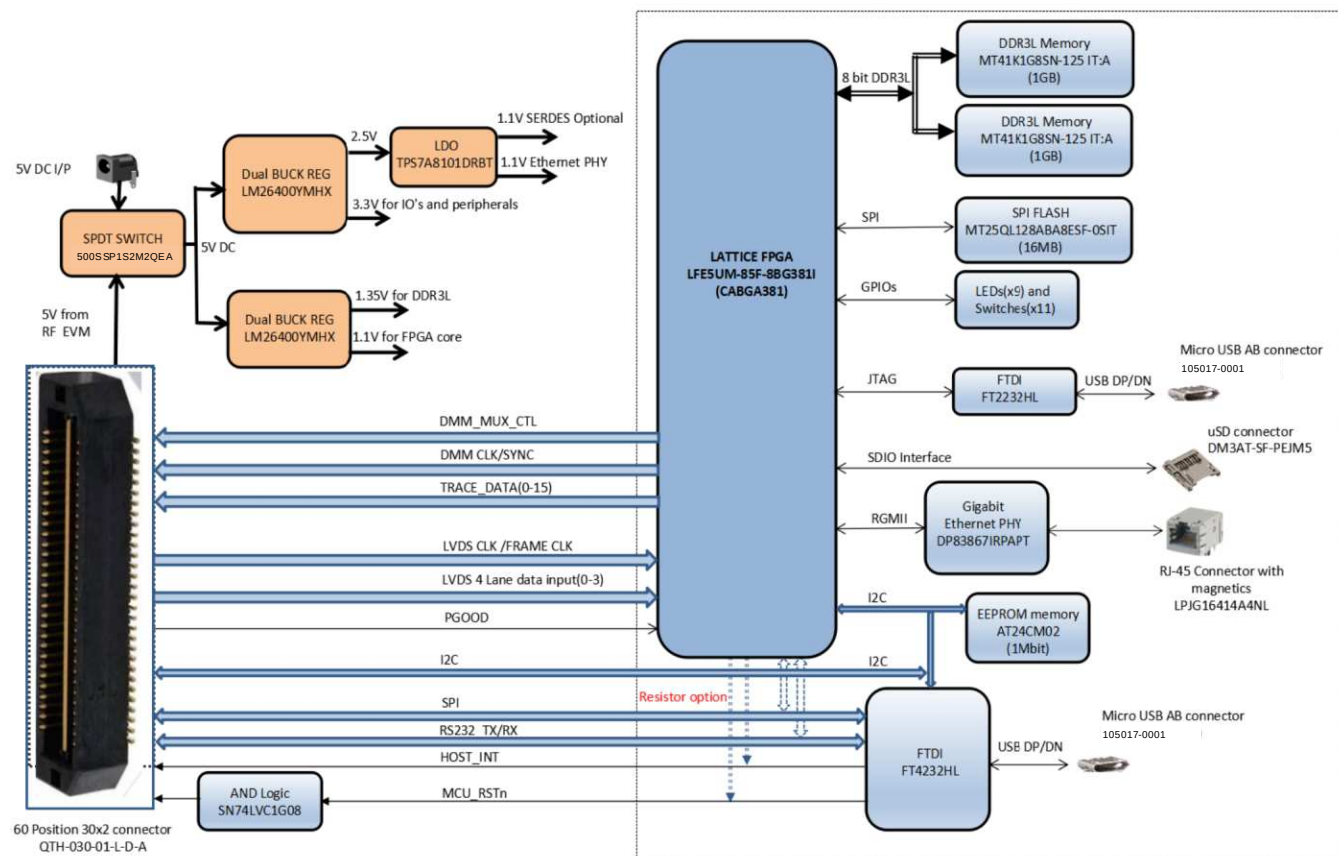
The following items are included with the DCA1000EVM kit. Unbox the board and identify various components and connectors as detailed in the user guide.

- DCA1000EVM Kit
- Ethernet cable
- Samtec coax micro ribbon cable
- 8GB microSD Card
- Spacers, screws, and washers

## 2 Hardware Specifications

### 2.1 Functional Block Diagram

The functional block diagram of the DCA1000EVM is shown in [Figure 1](#).



**Figure 1. DCA1000EVM Functional Block Diagram**

## 2.2 Board Dimensions

The DCA1000EVM dimensions are 90 mm × 102 mm × 1.6 mm. It is a 10-layer board fabricated with epoxy fiberglass FR4 grade material (UL94V-0 Certified).

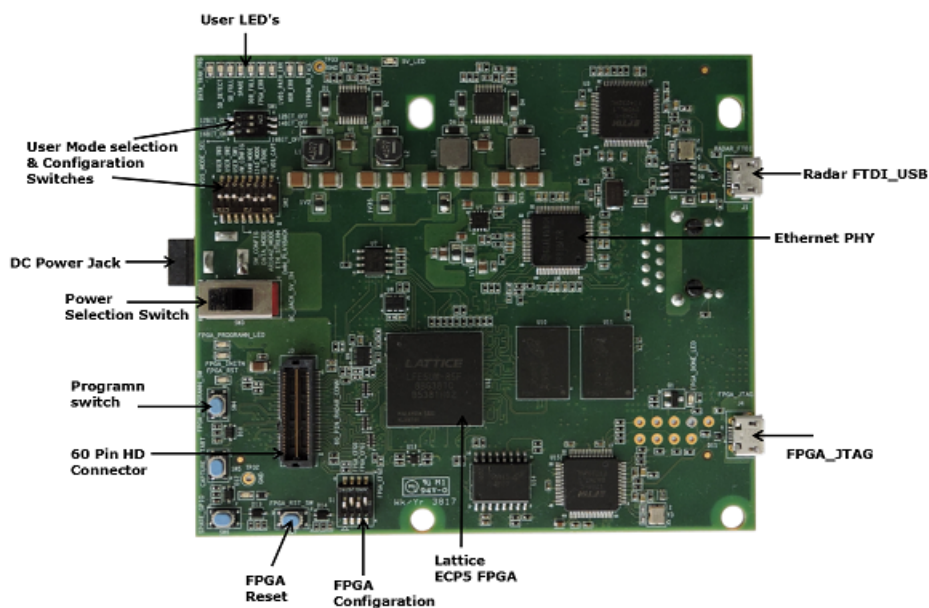


Figure 2. DCA1000EVM – Top View

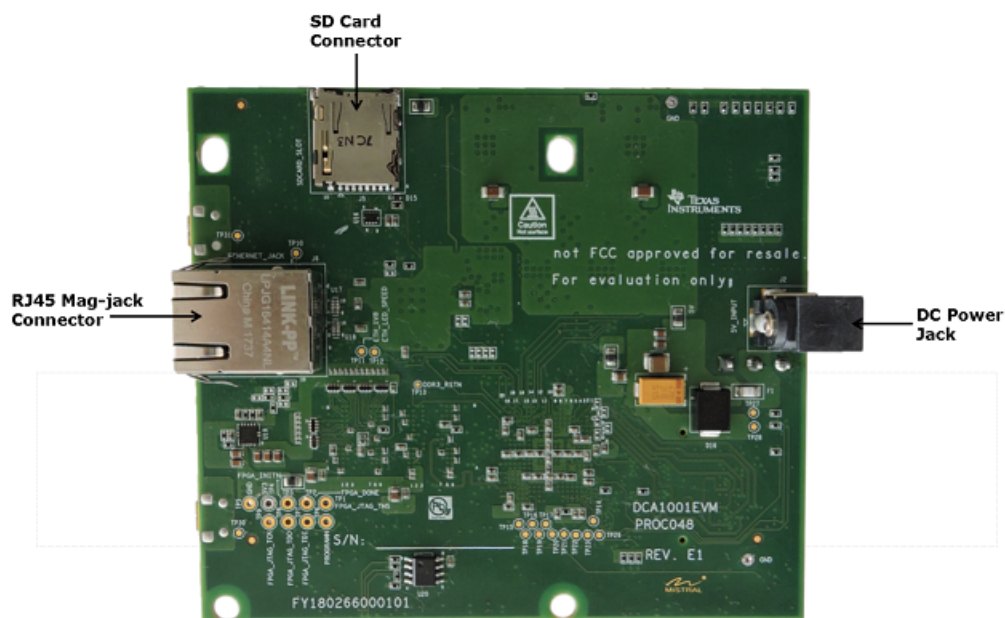


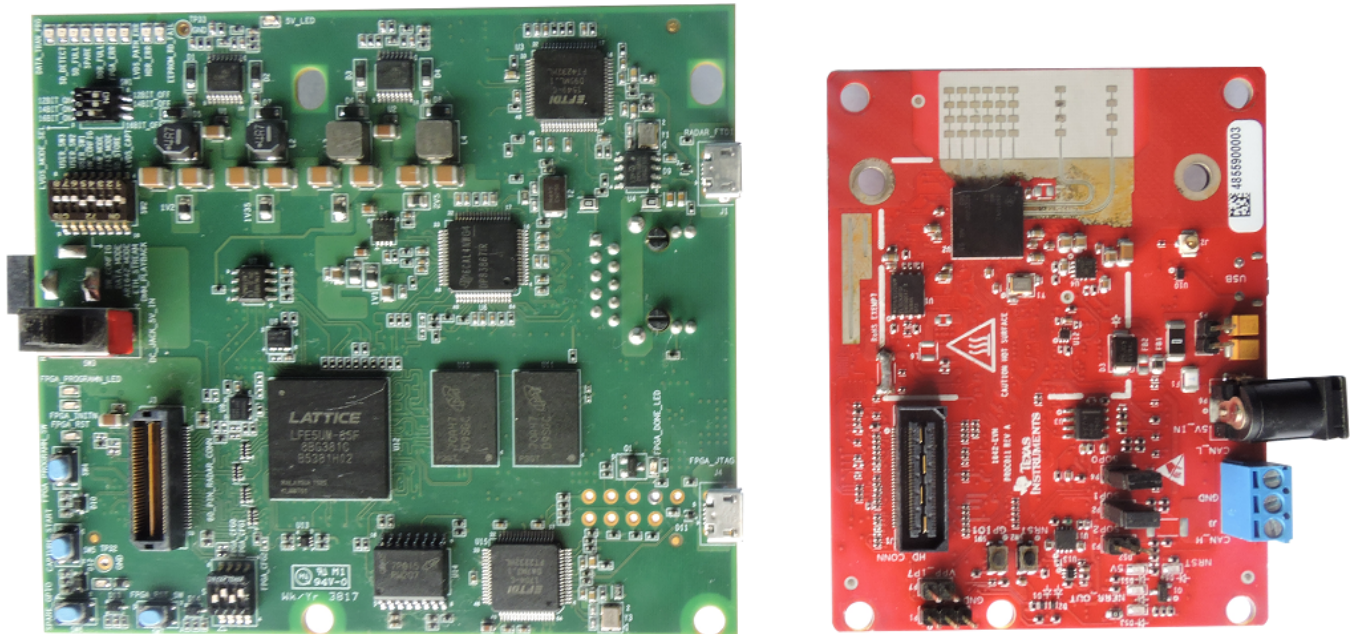
Figure 3. DCA1000EVM – Bottom View



## 2.3 DCA1000EVM and xWR1xxx EVM – Board to Board Connection

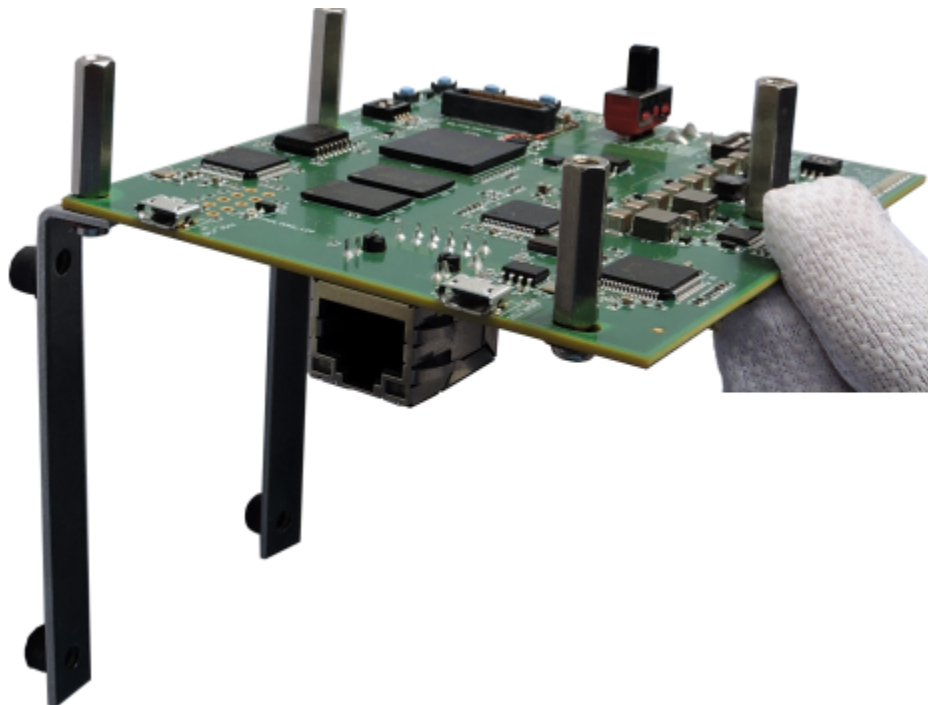
To enable easy measurements on the sensing objects on the horizontal plane, the xWR1xxx EVM (AWR1243BOOST/AWR1443BOOST/AWR1642BOOST) and DCA1000EVM can be mounted vertically by fixing the L-brackets. The following steps illustrate the assembly of the L-brackets, DCA1000EVM, and xWR1xxx EVM.

1. Align the DCA1000EVM and xWR1xxx EVM boards, as shown in [Figure 4](#).



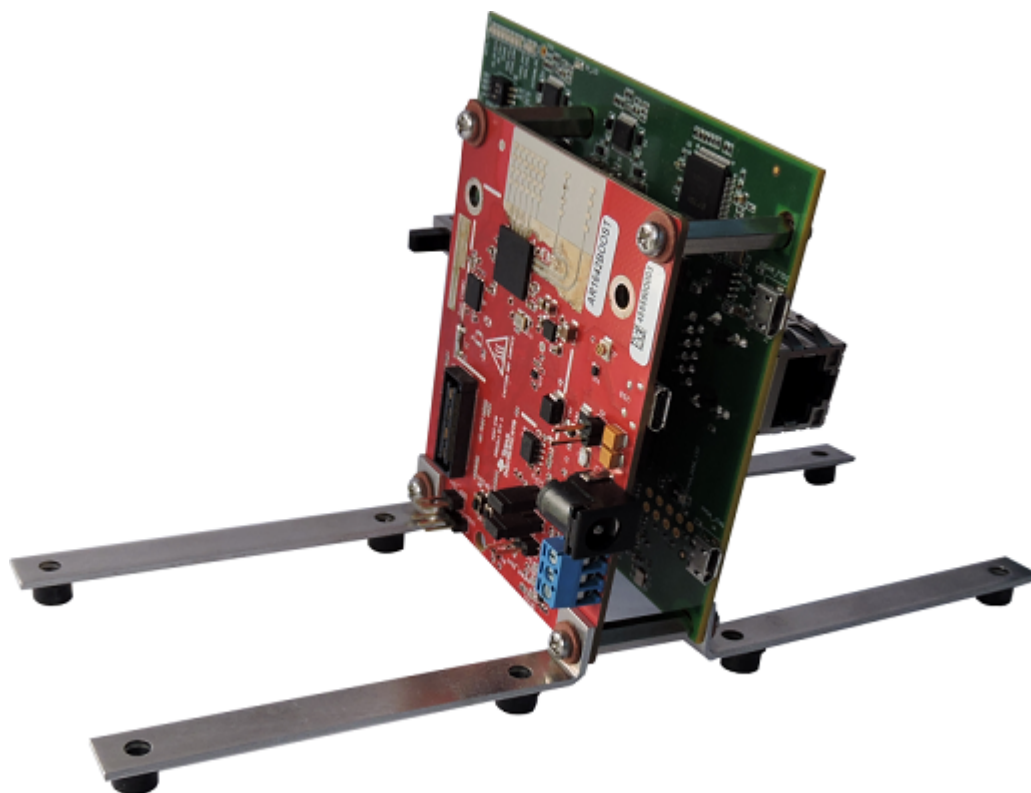
**Figure 4. DCA1000EVM and xWR1xxx EVM Board Alignment**

2. Connect the DCA1000EVM with screws, spacers, and stands, as shown in [Figure 5](#).



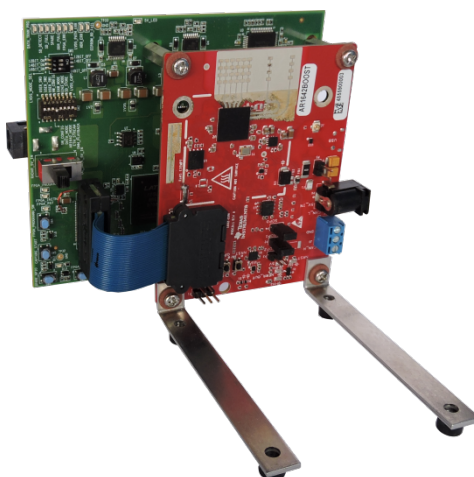
**Figure 5. Connect Screws and Spacers**

3. Connect the DCA1000EVM with the xWR1xxx EVM, as shown in [Figure 6](#).



**Figure 6. Connect Boards**

4. Connect the Samtec ribbon cable, as shown in [Figure 7](#) and [Figure 8](#).



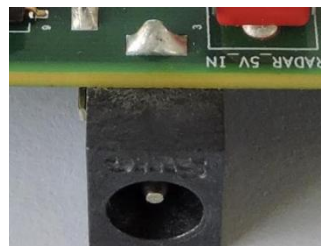
**Figure 7. Connect Ribbon Cable**



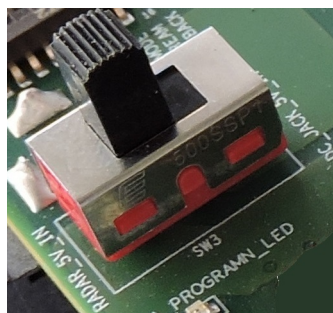
**Figure 8. Connect Ribbon Cable (Side)**

## 2.4 Power Connections

The DCA1000EVM is powered by 5-V power, either from a DC jack as shown in [Figure 9](#), or from a 60-pin high density connector of TI's xWR1xxx EVM mated to J3 of DCA1000EVM, as shown in [Figure 14](#). The power source is selected through the SPDT power switch ([Figure 10](#)) position, as shown in [Table 1](#). When the power is provided, the 5-V LED (LD10) glows, indicating that the board is powered on.



**Figure 9. Power Connector**



**Figure 10. Power Selection Switch**



**Table 1. Power Selection Switch Information**

| Reference | Usage                  | Description                                                                                                                                                                                                                                     |
|-----------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SW3.1     | Power selection switch | When the switch is closed to position 1, then the external 5-V power is given through a DC jack.                                                                                                                                                |
| SW3.3     | Power selection switch | When the switch is closed to position 3, then the 5-V power is derived from the xWR1xxx EVM through the 60-pin connectors.<br>Note: To get 5-V power from the xWR1xxx EVM, the R34 (0 $\Omega$ ) resistor should be mounted in the xWR1xxx EVM. |

## 2.5 Connectors

### 2.5.1 USB Connectors

The DCA1000EVM has two standard micro USB connectors. Connector J1 allows access to the xWR1xxx EVM through UART, SPI, or I2C through the FTDI chip, using the TI RADAR Studio application, and is as shown in [Figure 11](#). The mmWave Studio PC application can be used to configure the radar device. For more details on the mmWave Studio, refer to the following link: <http://www.ti.com/tool/MMWAVE-STUDIO>. The second micro USB connector J4 allows programming the FPGA through the FTDI chip through the JTAG interface, and is shown in [Figure 12](#).

[Table 2](#) provides the J1 connector pin information. [Table 3](#) provides the J4 connector pin information.

To detect the RADAR FTDI and emulate the SPI host bus, connect a micro USB cable to J1 on the DCA1000EVM.

**Table 2. J1 USB Connector Information**

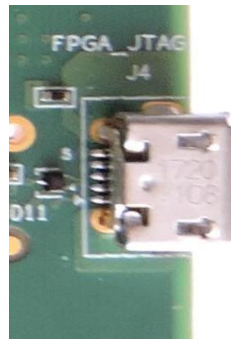
| Pin No | Signal Name  | Description              |
|--------|--------------|--------------------------|
| 1      | TP           | Test Point               |
| 2      | FTDI_USB1_DM | USB Data Signal Negative |
| 3      | FTDI_USB1_DP | USB Data Signal Positive |
| 4      | GND          | Ground                   |
| 5      | GND          | Ground                   |


**Figure 11. J1 USB Connector**

To configure the FPGA through JTAG, connect a micro USB cable to J4 on the DCA1000EVM.

**Table 3. J4 USB Connector Information**

| Pin No | Signal Name  | Description              |
|--------|--------------|--------------------------|
| 1      | TP           | Test Point               |
| 2      | FTDI_USB2_DM | USB Data Signal Negative |
| 3      | FTDI_USB2_DP | USB Data Signal Positive |
| 4      | GND          | Ground                   |
| 5      | GND          | Ground                   |


**Figure 12. J4 USB Connector**

## 2.5.2 Ethernet Jack

The DCA1000EVM supports a Gigabit Ethernet port to provide the connection to the network. The Ethernet port is interfaced to the LATTICE FPGA through the Ethernet PHY DP83867IRPAPT, and is used to stream the captured data over the network to the host PC. [Figure 13](#) shows the Ethernet RJ45 Mag-Jack connector, and [Table 4](#) provides the connector pin details.

**Table 4. J6 Ethernet Jack Information**

| Pin No | Signal Name | Description                                       |
|--------|-------------|---------------------------------------------------|
| 1      | GND         | Ground                                            |
| 2      | TP          | Test point                                        |
| 3      | ETH_D3P     | Gigabit Ethernet differential pair 3 positive     |
| 4      | ETH_D3N     | Gigabit Ethernet differential pair 3 negative     |
| 5      | ETH_D2P     | Gigabit Ethernet differential pair 2 positive     |
| 6      | ETH_D2N     | Gigabit Ethernet differential pair 2 negative     |
| 7      | ETH_D1P     | Gigabit Ethernet differential pair 1 positive     |
| 8      | ETH_D1N     | Gigabit Ethernet differential pair 1 negative     |
| 9      | ETH_D0P     | Gigabit Ethernet differential pair 0 positive     |
| 10     | ETH_D0N     | Gigabit Ethernet differential pair 0 negative     |
| 11     | 3V3         | Supply voltage for internal anode of activity LED |
| 12     | LED_ACTn    | Activity LED                                      |
| 13     | LED_LINKn   | Link LED                                          |
| 14     | 3V3         | Supply voltage for internal anode of link LED     |


**Figure 13. J6 Ethernet Jack**

### 2.5.3 60-Pin High Density (HD) Connector

The DCA1000EVM supports a 60-pin HD connector to interface with the xWR1xxx EVM. This connector provides the high-speed LVDS data, control signals (UART, SPI, I2C, RESET), and DMM interface signals. The DCA1000EVM captures data from the xWR1xxx EVM over a 4-lane LVDS interface and playbacks the data from the host PC to the xWR1xxx EVM over a 16-bit DMM/trace interface through the 60-pin connector. Control signals on the 60-pin HD connector are used to access the xWR1xxx EVM through the FTDI chip. [Figure 14](#) shows the HD connector, and [Table 5](#) provides the connector pin details.

**Table 5. J3 High Density Interface Connector Information**

| Pin No | Signal Name              | Description                         |
|--------|--------------------------|-------------------------------------|
| 1      | VCC5V_CONN               | 5-V supply voltage from xWR1xxx EVM |
| 2      | VCC5V_CONN               | 5-V supply voltage from xWR1xxx EVM |
| 3      | VCC5V_CONN               | 5-V supply voltage from xWR1xxx EVM |
| 4      | NC                       | No connect                          |
| 5      | NC                       | No connect                          |
| 6      | NC                       | No connect                          |
| 7      | Conn_SPI_CS <sub>n</sub> | SPI chip select                     |
| 8      | NC                       | No connect                          |
| 9      | Conn_SPI_CLK             | SPI clock                           |
| 10     | HOST_INT                 | Host interrupt from FTDI            |
| 11     | Conn_SPI_MOSI            | SPI master output slave input       |
| 12     | Conn_SPI_MISO            | SPI master input slave output       |
| 13     | PGOOD                    | PGOOD input from xWR1xxx EVM        |
| 14     | NC                       | No connect                          |
| 15     | DMM_CLK                  | DMM clock                           |
| 16     | DMM_MUX_CTL              | DMM control                         |
| 17     | DMM_SYNC                 | DMM synchronization                 |
| 18     | GND                      | Ground                              |
| 19     | DP0                      | DMM Data0                           |
| 20     | LVDS_VALIDP              | LVDS valid signal positive          |
| 21     | DP1                      | DMM Data1                           |
| 22     | LVDS_VALIDM              | LVDS valid signal negative          |
| 23     | DP2                      | DMM Data2                           |
| 24     | GND                      | Ground                              |
| 25     | DP3                      | DMM Data3                           |
| 26     | LVDS_FRCLKP              | LVDS frame clock signal positive    |
| 27     | DP4                      | DMM Data4                           |
| 28     | LVDS_FRCLKN              | LVDS frame clock signal negative    |

**Table 5. J3 High Density Interface Connector Information (continued)**

| Pin No | Signal Name | Description                |
|--------|-------------|----------------------------|
| 29     | DP5         | DMM Data5                  |
| 30     | GND         | Ground                     |
| 31     | DP6         | DMM Data6                  |
| 32     | LVDS_3P     | LVDS data pair 3 positive  |
| 33     | DP7         | DMM Data7                  |
| 34     | LVDS_3M     | LVDS data pair 3 negative  |
| 35     | DP8         | DMM Data8                  |
| 36     | GND         | Ground                     |
| 37     | DP9         | DMM Data9                  |
| 38     | LVDS_2P     | LVDS data pair 2 positive  |
| 39     | DP10        | DMM Data10                 |
| 40     | LVDS_2M     | LVDS data pair 2 negative  |
| 41     | DP11        | DMM Data11                 |
| 42     | GND         | Ground                     |
| 43     | DP12        | DMM Data12                 |
| 44     | LVDS_CLKP   | LVDS clock pair positive   |
| 45     | DP13        | DMM Data13                 |
| 46     | LVDS_CLKM   | LVDS clock pair negative   |
| 47     | DP14        | DMM Data14                 |
| 48     | GND         | Ground                     |
| 49     | DP15        | DMM Data15                 |
| 50     | LVDS_1P     | LVDS data pair 1 positive  |
| 51     | SDA         | Serial Data                |
| 52     | LVDS_1M     | LVDS data pair 1 negative  |
| 53     | SCL         | Serial clock               |
| 54     | GND         | Ground                     |
| 55     | RS232RX     | UART data receive          |
| 56     | LVDS_0P     | LVDS data pair 0 positive  |
| 57     | RS232TX     | UART data transmit         |
| 58     | LVDS_0M     | LVDS data pair 0 negative  |
| 59     | MCU_RSTn    | Reset input to xWR1xxx EVM |
| 60     | GND         | Ground                     |


**Figure 14. J3 High Density Interface Connector**

## 2.6 Switches, Push Buttons, and LEDs

### 2.6.1 Switches

The DCA1000EVM supports a 1×3 DIP switch (as explained in [Table 7](#)) and a 1×8 DIP switch (as explained in [Table 8](#)) for mode selection and other functions. In switch SW2, Pin5 indicates that the DCA1000EVM is configured either by a hardware switch-based configuration, or by a software-based configuration, as listed in [Table 6](#).

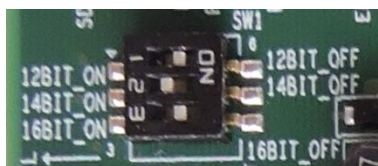
**Table 6. Switch2.5 Functionality Information**

| Reference | Usage         | Description                                                                                                                                                                                              |
|-----------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SW2.5     | CONFIG_VIA_HW | When positioned at 5 (Pin5), the DCA1000EVM is configured as per the settings of the hardware DIP switch SW2.<br>In hardware configuration mode, software configurations are ignored                     |
|           | CONFIG_VIA_SW | When positioned at the other side (Pin12), the DCA1000EVM is configured through the software configuration commands over Ethernet.<br>In software configuration mode, hardware DIP settings are ignored. |

When the DCA1000EVM is configured in hardware switch configuration mode, use the following switch settings for different modes of data capture.

**Table 7. Switch1 Functionality Information**

| LVDS bit Setting Switch | SW1 (3 Position Switch) |     |     | Description                                        |
|-------------------------|-------------------------|-----|-----|----------------------------------------------------|
|                         | 1                       | 2   | 3   |                                                    |
| LVDS 12-bit mode        | ON                      | OFF | OFF | LVDS 12 bit data is captured from the xWR1xxx EVM. |
| LVDS 14-bit mode        | OFF                     | ON  | OFF | LVDS 14 bit data is captured from the xWR1xxx EVM. |
| LVDS 16-bit mode        | OFF                     | OFF | ON  | LVDS 16 bit data is captured from the xWR1xxx EVM. |



**Figure 15. Mode Configuration\_1**

**Table 8. Switch2 Functionality Information**

| Reference | Usage        | Description                                                                                                                                                                                                                    |
|-----------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SW2.1     | LVDS_CAPTURE | When positioned at 1(Pin1), the DCA1000EVM is in capture mode. In this mode, data is received over LVDS and streamed through Ethernet to the PC.                                                                               |
|           | DMM_PLAYBACK | When positioned at the other side (Pin16), the DCA1000EVM is in play back mode.                                                                                                                                                |
| SW2.2     | SD_STORE     | When positioned at 2 (Pin2), the DCA1000EVM is in capture mode, and data is saved into the SD card.                                                                                                                            |
|           | ETH_STREAM   | When positioned at the other side (Pin15), the DCA1000EVM is in capture mode, and data is streamed over the network to the host PC.                                                                                            |
| SW2.3     | 1243_MODE    | When positioned at 3 (Pin3), the DCA1000EVM captures 4-lane LVDS data.                                                                                                                                                         |
|           | AR1642_MODE  | When positioned at the other side (Pin14), the DCA1000EVM captures 2-lane LVDS data.                                                                                                                                           |
| SW2.4     | RAW_MODE     | When positioned at 4 (Pin4), the DCA1000EVM is in raw mode. In this mode, all LVDS data is captured as it is.                                                                                                                  |
|           | DATA_MODE    | When positioned at the other side (Pin13), the DCA1000EVM is in data separated mode. In this mode, the user can add specific headers to different data types, and FPGA separates out different data types based on the header. |



**Table 8. Switch2 Functionality Information (continued)**

| Reference | Usage                                    | Description                                                                                                                        |
|-----------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| SW2.6     | USER_SW1<br>(Using EEPROM configuration) | When positioned at 6 (Pin6), the Ethernet configuration data is read from EEPROM, and the same is used for Ethernet communication. |
|           | GND<br>(Using FPGA configuration)        | When positioned at the other side (Pin11), the Ethernet configuration data hardcoded in FPGA is used for Ethernet communication.   |
| SW2.7     | USER_SW2                                 | Future use                                                                                                                         |
|           | GND                                      |                                                                                                                                    |
| SW2.8     | USER_SW3                                 | Future use                                                                                                                         |
|           | GND                                      |                                                                                                                                    |

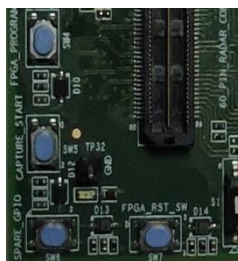

**Figure 16. Mode Configuration\_2**

## 2.6.2 Push Buttons

The DCA1000EVM supports four push buttons for FPGA reset, data capture start, and other basic functions listed in [Table 9](#).

**Table 9. Push Button Functionality Information**

| Reference | Usage         | Description                                                                                                                                                                        |
|-----------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SW4       | PROGRAMN      | Whenever SPI flash is re-flashed or reprogrammed, press the PROGRAMN (SW4) button to program the FPGA from SPI flash.<br>Note: A binary file should be present in the SPI flash.   |
| SW5       | Capture start | When the DCA1000EVM is in hardware configuration mode, press the capture start (SW5) button to initiate the data transfer. Press the button (SW5) again to stop the data transfer. |
| SW6       | SPARE_GPIO_SW | Future use                                                                                                                                                                         |
| SW7       | FPGA_RST      | Press the FPGA_RST button to reset the FPGA.                                                                                                                                       |


**Figure 17. Push Button**

## 2.6.3 LEDs

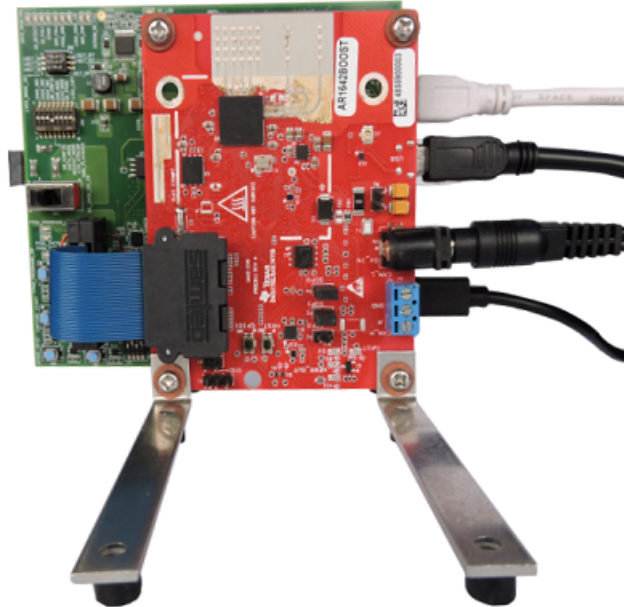
The DCA1000EVM supports LEDs for user indications, as listed in [Table 10](#).

**Table 10. LED Functionality Information**

| Reference | Usage                | Description                                                                                                 | Color |
|-----------|----------------------|-------------------------------------------------------------------------------------------------------------|-------|
| LD1       | DATA_TRANS_PROG_LED1 | This LED indicates the progress of data transfer, either through the network or saving into the SD card.    | Green |
| LD2       | SD_DETECT_LED5       | When the SD card is mounted into the DCA1000EVM, this LED indicates whether the SD card is detected or not. | Green |
| LD3       | SD_FULL_LED0         | This LED indicates the full condition of the SD card.                                                       | Red   |
| LD4       | SPARE_LED8           | Future use                                                                                                  | Green |
| LD5       | FPGA_ERR_LED2        | When LVDS data is coming at high rate, the internal FPGA buffer gets full. In that case, this LED glows.    | Red   |
| LD6       | DDR_FULL_LED7        | This LED indicates the full condition of the 2GB DDR3 memory.                                               | Red   |
| LD7       | LVDS_PATH_ERR_LED3   | This LED indicates whether the xWR1xxx EVM is sending LVDS data or not.                                     | Red   |
| LD8       | HEADER_ERR_LED4      | This LED indicates the failure of the header when the DCA1000EVM is in data separated mode.                 | Red   |
| LD9       | EEPROM_RD_FAIL_LED6  | This LED indicates an EEPROM read access failure.                                                           | Red   |
| LD10      | 5V                   | This LED indicates the presence of the 5-V supply in the DCA1000EVM.                                        | Green |
| LD11      | PROGRAMN             | This LED indicates assertion of the program to FPGA when the PROGRAMN button is pressed.                    | Red   |
| LD12      | INITN                | This LED indicates that the FPGA is ready to be configured.                                                 | Red   |
| LD13      | FPGA_RST             | This LED indicates the assertion of the FPGA reset.                                                         | Red   |
| LD14      | DONE                 | When FPGA is programmed from SPI flash, this LED glows.                                                     | Green |
| LD15      | FTDI_JTAG_ACT        | This LED indicates activity on the FPGA JTAG connectivity.                                                  | Green |


**Figure 18. User LEDs**

### 3 System Setup



**Figure 19. Board Setup**

#### 3.1 Prerequisites

- The DCA1000EVM should be connected to the host PC through an Ethernet cable for the data transfer process.
- The DCA1000EVM should be connected to a PC through a USB cable (J1-Radar FTDI) for configuring the xWR1xxx EVM if the mmWave Studio is used to configure the radar device. If an embedded application is used to configure the xWR1xxx EVM, then this is not required.
- The DCA1000EVM should be connected to TI's xWR1xxx EVM through a 60-pin HD connector by using a 60-pin Samtec ribbon cable.
- The DCA1000EVM power input should be connected from either a DC jack or TI's xWR1xxx EVM power output (from a 60-pin HD connector) by selecting the SW3 switch position.
- The xWR1xxx EVM should be connected to a PC through a USB cable (J8) for the RS232 connection in the Radar Studio interface.
- HOST PC configuration:
  - Intel Core i5 2.5 GHz or above
  - 8 GB RAM or above
  - Windows 10 Pro operating system
  - Static IP Address – 192.168.33.30
  - 1-Gbps Ethernet port
- Pre-installed mmWave Studio in the [mmWave Studio user guide](#).
- The DCA1000EVM should be connected to a PC through a USB cable (J4-FPGA JTAG) for programming the FPGA (when the card is in programming mode).

## 4 DCA1000EVM Ethernet Configuration Data

The DCA1000EVM stores the default Ethernet configuration data (such as DCA1000EVM IP, DCA1000EVM MAC, System IP, Configuration port, and Data port) in FPGA internal registers.

The default Ethernet configuration data stored in the FPGA internal registers are listed in [Table 11](#).

**Table 11. Table 11 Default Value Stored in FPGA Registers**

| Parameter                        | Default Value (Decimal) |
|----------------------------------|-------------------------|
| FPGA IP                          | 192.168.33.180          |
| FPGA MAC                         | 12-34-56-78-90-12       |
| System IP                        | 192.168.33.30           |
| Configuration port               | 4096                    |
| ADC/CP_ADC/ADC_CP/CP_ADC_CQ data | 4098                    |
| CP data                          | 4099                    |
| CQ data                          | 4100                    |
| R4F debug data                   | 4101                    |
| DSP debug data                   | 4102                    |

The DCA1000EVM has option of loading Ethernet configuration data from EEPROM or the FPGA internal register, based on SW2.6.

The Ethernet configuration data is loaded from the EEPROM when SW2 positioned at 6 (pin6); else, FPGA register values are loaded.

Loading the Ethernet configuration data from EEPROM:

- When SW2 is positioned at 6 (pin6), the DCA1000EVM loads the Ethernet configuration data from EEPROM.
- In general, the EEPROM is blank when the DCA1000EVM comes out of the factory. The EEPROM remains blank until the updated EEPROM command is given to the DCA1000EVM through the Ethernet port.
- When the updated EEPROM command is given to the DCA1000EVM, the Ethernet configuration data are stored into the EEPROM along with the header and footer.
- The DCA1000EVM requires a power cycle/hardware reset to load the updated Ethernet configuration data from the EEPROM.
- During the power ON/after hardware reset, the DCA1000EVM FPGA loads the Ethernet configuration data from the EEPROM and checks for the header and footer. If the header and footer are found, the data loaded from the EEPROM is used for Ethernet communication; else, the FPGA uses the default values stored in the internal registers.
- The Ethernet configuration data updated in the EEPROM should be used in the Ethernet packets which are going as commands to the DCA1000EVM, for further communication. Otherwise, the DCA1000EVM will not respond to the command.

Loading the Ethernet configuration data from the FPGA internal registers:

- When SW2 is positioned at (pin11), the DCA1000EVM loads the default Ethernet configuration data from the FPGA internal registers, and the same is used for Ethernet communication.

---

**NOTE:** For data separated mode, the DCA1000EVM FPGA receives only one data port number from the updated EEPROM command. This port is considered as an ADC data port, and other data type's port number is incremented by one.

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**NOTE:** Whenever Switch 2.6 position is changed from one option to another option, the DCA1000EVM requires a power cycle and hardware reset.

---

## 5 DCA1000EVM Command and Data Format

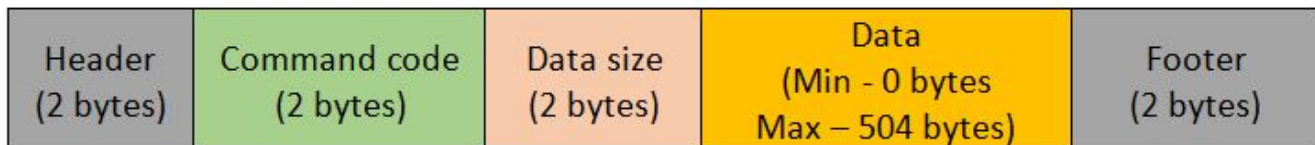
The DCA1000EVM follows UDP protocol and supports 14 predefined commands. The configuration and status of the DCA1000EVM are communicated through the configuration port. The data port is used to transfer raw mode/data separated mode data.

### 5.1 DCA1000EVM Command Format

The DCA1000EVM receives a command from the host through the configuration port, and sends back the response in the same port.

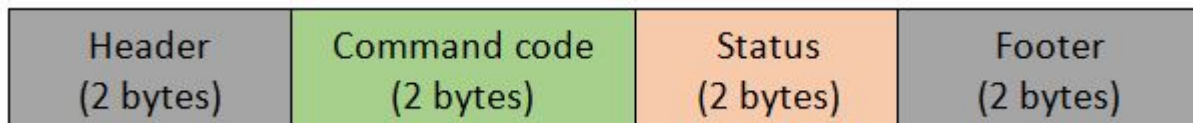
The Command Request protocol consists of the following:

- Header
  - a. Header
  - b. Command code
  - c. Data size
- Data
- Footer



The Command Response protocol consists of the following:

- Header
  - a. Header
  - b. Command code
- Status
- Footer



Commands supported in the DCA1000EVM are listed in [Table 12](#).

**Table 12. Supported Commands**

| Command                          | Command Code |
|----------------------------------|--------------|
| RESET_FPGA_CMD_CODE              | 0x01         |
| RESET_AR_DEV_CMD_CODE            | 0x02         |
| CONFIG_FPGA_GEN_CMD_CODE         | 0x03         |
| CONFIG_EEPROM_CMD_CODE           | 0x04         |
| RECORD_START_CMD_CODE            | 0x05         |
| RECORD_STOP_CMD_CODE             | 0x06         |
| PLAYBACK_START_CMD_CODE          | 0x07         |
| PLAYBACK_STOP_CMD_CODE           | 0x08         |
| SYSTEM_CONNECT_CMD_CODE          | 0x09         |
| SYSTEM_ERROR_CMD_CODE            | 0x0A         |
| CONFIG_PACKET_DATA_CMD_CODE      | 0x0B         |
| CONFIG_DATA_MODE_AR_DEV_CMD_CODE | 0x0C         |

**Table 12. Supported Commands (continued)**

| Command                     | Command Code |
|-----------------------------|--------------|
| INIT_FPGA_PLAYBACK_CMD_CODE | 0x0D         |
| READ_FPGA_VERSION_CMD_CODE  | 0x0E         |

Table 13 and Table 14 show examples using RESET\_FPGA\_CMD\_CODE.

**Table 13. Reset FPGA Command Request From Host**

| Name         | Data Type | Number of Bytes | Default Value | Min Value | Max Value | Description                          |
|--------------|-----------|-----------------|---------------|-----------|-----------|--------------------------------------|
| Header       | UINT16    | 2               | 0xA55A        | -         | -         | 0xA55A always. Start bits of packet. |
| Command code | UINT16    | 2               | 0x01          | -         | -         | Command code                         |
| Size         | UINT16    | 2               | 0             | -         | -         | Data size                            |
| Footer       | UINT16    | 2               | 0xEEAA        | -         | -         | 0xEEAA always. Stop bits of packet.  |

**Table 14. Command Response From the DCA1000EVM**

| Name         | Data Type | Number of Bytes | Default Value | Min Value | Max Value | Description                          |
|--------------|-----------|-----------------|---------------|-----------|-----------|--------------------------------------|
| Header       | UINT16    | 2               | 0xA55A        | -         | -         | 0xA55A always. Start bits of packet. |
| Command code | UINT16    | 2               | 0x01          | -         | -         | Command code                         |
| Status       | UINT16    | 2               | 0             | 0         | 1         | 0 – Success<br>1 – Failure           |
| Footer       | UINT16    | 2               | 0xEEAA        | -         | -         | 0xEEAA always. Stop bits of packet.  |

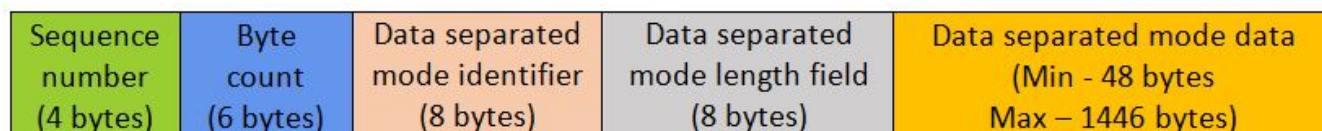
## 5.2 DCA1000EVM Data Format

The DCA1000EVM sends raw mode/data separated mode data to the host through the data port.

- Raw mode data format: the DCA1000EVM sends raw mode data in the following format.



- Data separated mode data format: the DCA1000EVM sends data separated mode data in the following format.

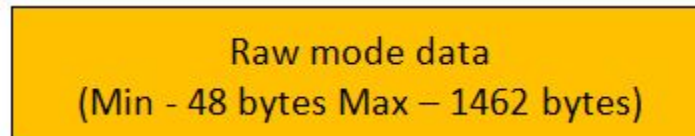




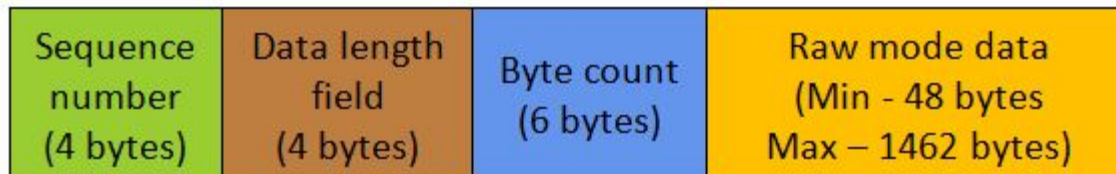
### 5.3 Stored File Format

Raw mode data format: in raw mode, the file is stored in the following format:

- Without sequence number:

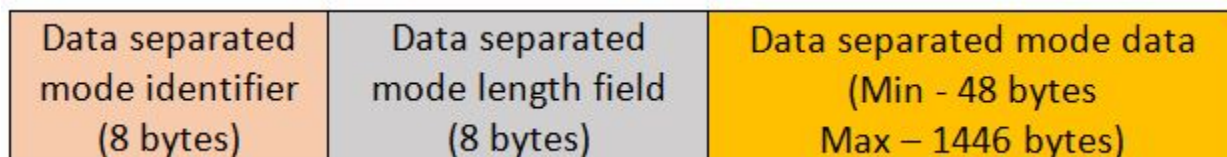


- With sequence number:

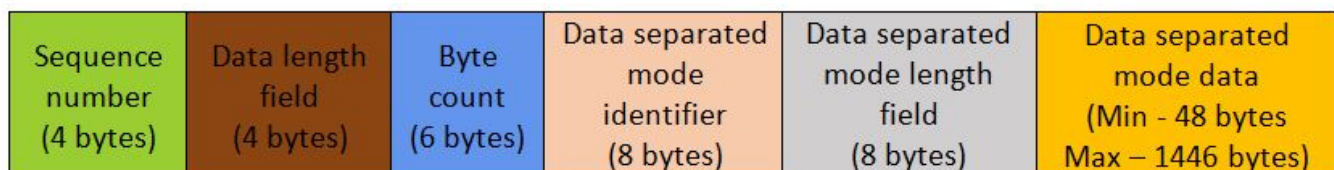


Data separated mode data format: in data separated mode, the file is stored in the following format:

- Without sequence number:



- With sequence number:



## 6 Use Case: LVDS Over Ethernet Streaming

LVDS over Ethernet streaming can be used in two modes of operations: raw mode and data separated mode. Both can be accessed either through the hardware switch configuration mode or software GUI configuration mode by selecting the switch (SW2.5).

### 6.1 Raw Mode

In Raw mode, the FPGA records all the data appearing on the LVDS lines as is. The data sent over the LVDS lanes is controlled by either the configuration from the mmwave studio, or the user application running on the radar device.

### 6.2 Data Separated Mode

Data separated mode features supports for receiving different data types with unique identifiers, along with length field information from the xWR1xxx EVM over the LVDS interface, and streamed over the Ethernet interface in different ports. This mode of capture supports specific identifiers, as listed in [Table 15](#). Each data type has a 64-bit identifier and 64-bit length field. In the length field, the actual length information is only 24 bits (the first 12 bits are taken from the first sample and the next 12 bits from the second sample). The remaining 40 bits are reserved. The user application on the radar device must ensure that these headers are added on the LVDS data in the exact format the FPGA expects.

**Table 15. Data Separated Mode Data, Identifier, Length Information**

| Data type      | Identifier (64 bits) | Length Field (64 bits)                | Actual Data |
|----------------|----------------------|---------------------------------------|-------------|
| ADC Data       | 0x0ADC0CDA0ADC0CDA   | Actual length of data is only 24 bits | Valid Data  |
| CP Data        | 0x0CC909CC0CC909CC   |                                       |             |
| CQ Data        | 0x0CC808CC0CC808CC   |                                       |             |
| R4F Debug Data | 0x084F0F48084F0F48   |                                       |             |
| DSP Debug Data | 0x0C67076C0C67076C   |                                       |             |
| CP_ADC Data    | 0x0ADD0DDA0ADD0DDA   |                                       |             |
| ADC_CP Data    | 0x0ADE0EDA0ADE0EDA   |                                       |             |
| CP_ADC_CQ Data | 0x0ADF0FDA0ADF0FDA   |                                       |             |

Each data is streamed in different Ethernet ports through an Ethernet interface, as listed in [Table 16](#).

**Table 16. Data Type and Port for Data Separated Mode**

| Data type                                       | Port |
|-------------------------------------------------|------|
| ADC Data/CP_ADC Data/ADC_CP Data/CP_ADC_CQ Data | 4098 |
| CP Data                                         | 4099 |
| CQ Data                                         | 4100 |
| R4F Debug Data                                  | 4101 |
| DSP Debug Data                                  | 4102 |

## 7 Errors

**Table 17. Errors**

| Error indication       | Description                                                                                                                                                                                                     | Troubleshooting                                                                                                                                                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Out of sequence packet | The host PC might miss some Ethernet packets or receive jumbled packets from the DCA1000EVM. This may be due to either Ethernet fast streaming, UDP protocol, host PC configuration, or the network card (NIC). | Change the Ethernet packet delay options to avoid out of sequence. When increasing Ethernet packet delay, streaming throughput is decreased due to slower data capturing.                                                                                 |
| FPGA_ERR_LED           | This LED indicates an FPGA internal LVDS buffer overflow error. This error occurs when the LVDS data rate is faster than the Ethernet data rate.                                                                | Change the periodicity in the radar studio sensorconfig settings to avoid LVDS buffer overflow conditions.                                                                                                                                                |
| DDR_FULL_LED           | This LED indicates a DDR3 full error. This condition occurs when the LVDS data rate is faster than the Ethernet data rate                                                                                       | Change the periodicity in the radar studio sensorconfig settings to avoid DDR3 full conditions.                                                                                                                                                           |
| LVDS_PATH_ERR_LED      | This LED indicates NO LVDS DATA received from the radar EVM within a timeout period.                                                                                                                            | This occurs if the xWR1xxx EVM is not sending LVDS data to the DCA1000EVM. Check whether the radar EVM is configured and enabled to send data. Check that the cabling assembly is correct between the xWR1xxx EVM and the DCA1000EVM.                     |
| HEADER_ERR_LED         | This LED glows when NO HEADER is found in the LVDS data.                                                                                                                                                        | This occurs if the xWR1xxx EVM is not sending LVDS data to the DCA1000EVM. Check whether the radar EVM is configured and enabled to send data separated mode data. Check that the cabling assembly is correct between the xWR1xxx EVM and the DCA1000EVM. |
| EEPROM_RD_FAIL_LED     | This LED indicates EEPROM read failure status. This error can happen if any problem exists with EEPROM hardware connectivity or EEPROM addressing problem.                                                      | Power cycle the DCA1000EVM. Check EEPROM connectivity and address lines on hardware.                                                                                                                                                                      |

## 8 Troubleshooting

- To confirm Ethernet connectivity between the DCA1000EVM and host PC, use the Wireshark tool to check the Ethernet packet transfer.
- Ensure that UDP is enabled in the host PC for UDP data transmission and reception.
- When providing a hardware reset or power cycle on the board, before connecting the mmWave Studio with the DCA1000EVM, check whether the host PC Ethernet connectivity is established or not.
- Ensure that the Samtec cable is properly connected between the xWR1xxx EVM and DCA1000EVM to avoid signal discontinuity.
- The default Ethernet delay is pre-programmed as 25  $\mu$ s in both the FPGA and software. The user can configure the delay from 5  $\mu$ s to 500  $\mu$ s, depending on the host PC configuration and capabilities, to avoid packet loss or packets out of sequence. Lower delay allows for higher bandwidth while increasing the probability of UDP packet drops at the PC end. Refer to [Table 19](#) for the relationship between the packet delay and ethernet throughput.
- When the LVDS data rate is changed in Radar Studio, the DCA1000EVM requires a hardware/software reset to lock the PLL with the incoming clock in the FPGA.
- To update the FPGA image, follow the procedure explained in [Section 9](#).

## 9 FPGA Programming

The DCA1000EVM supports four DIP switch (S1) options for the FPGA programming configuration, as listed in [Table 18](#).

**Table 18. FPGA Configuration Mode**

| Configuration Mode | S1[4:2] | S1.4             | S1.3             | S1.2             |
|--------------------|---------|------------------|------------------|------------------|
| Master SPI         | 010     | Pin5 (FPGA_CFG2) | Pin6 (FPGA_CFG1) | Pin7 (FPGA_CFG0) |



**Figure 20. FPGA Config Switch**

For FPGA programming, install the Lattice Diamond Programmer standalone tool in the PC, as per the following steps.

1. Download the Lattice Diamond programmer tool, Windows version (Programmer Standalone 3.10 64-bit for Windows) from the following link: <http://www.latticesemi.com/programmer>.
2. A new folder named Iscc will be created at the path of installation.

Post installation steps:

1. Connect the USB cable to the FPGA JTAG USB port (J4), and power on the board.
2. The FTDI cable driver is installed automatically. Otherwise, manually select the cable driver from the installation path.
3. The device manager will recognize two COM ports, as shown in [Figure 21](#).

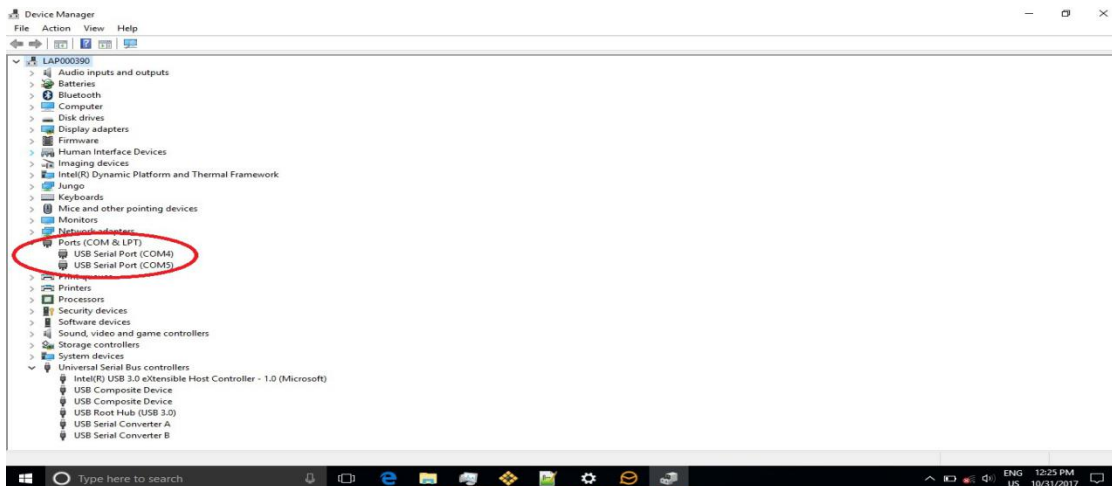


Figure 21. Device Manager

## 9.1 Hardware Setup

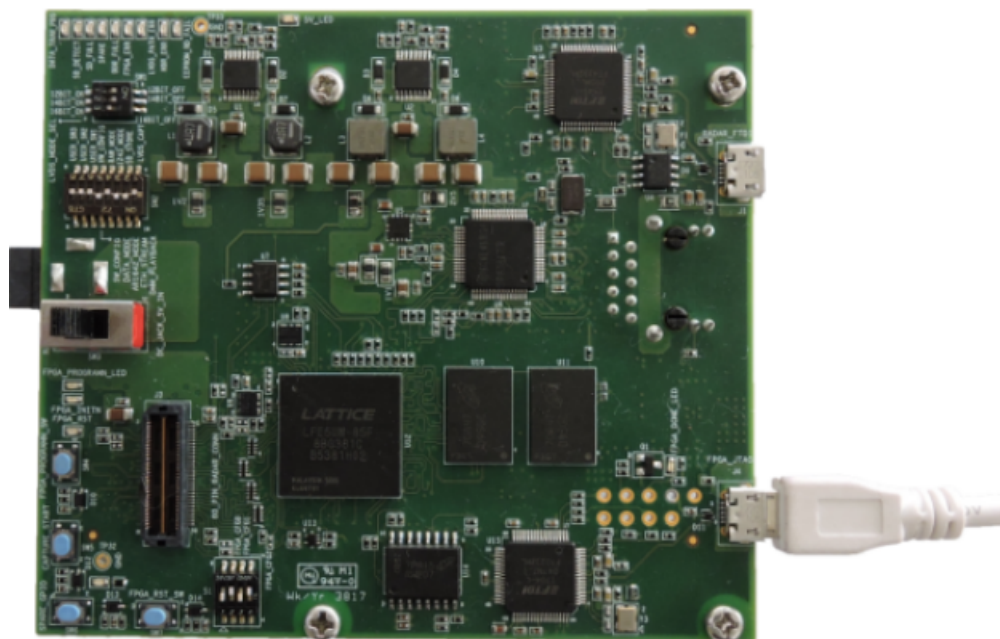
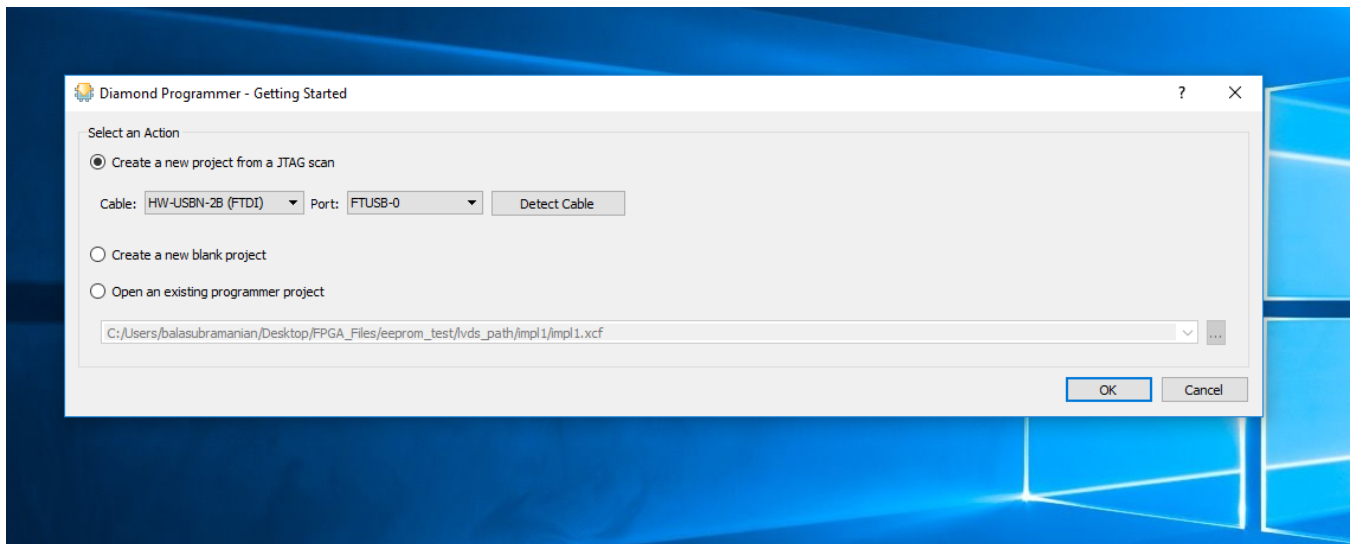


Figure 22. Hardware Setup

## 9.2 FPGA - SPI Flash Programming Mode

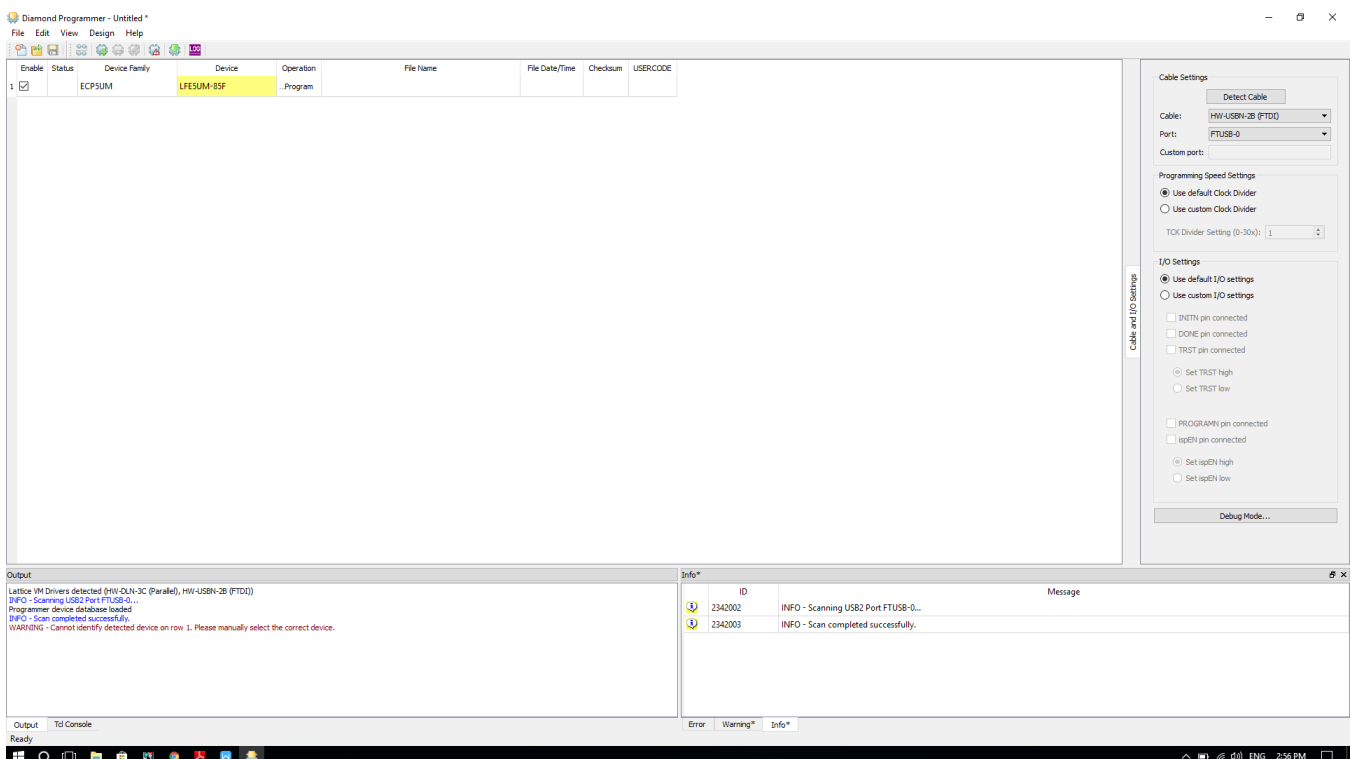
The FPGA image can be flashed in the onboard SPI flash once. This image is loaded to the FPGA each time the FPGA boots up. By default, the FPGA image for data capture would already be flashed in DCA1000EVM. The user must only perform this step if they need to update to a newer version of image, or re-flash the image due to some issues.

1. Open the Lattice Diamond programmer tool from the Start menu and select Detect Cable. Click OK as shown in [Figure 23](#).



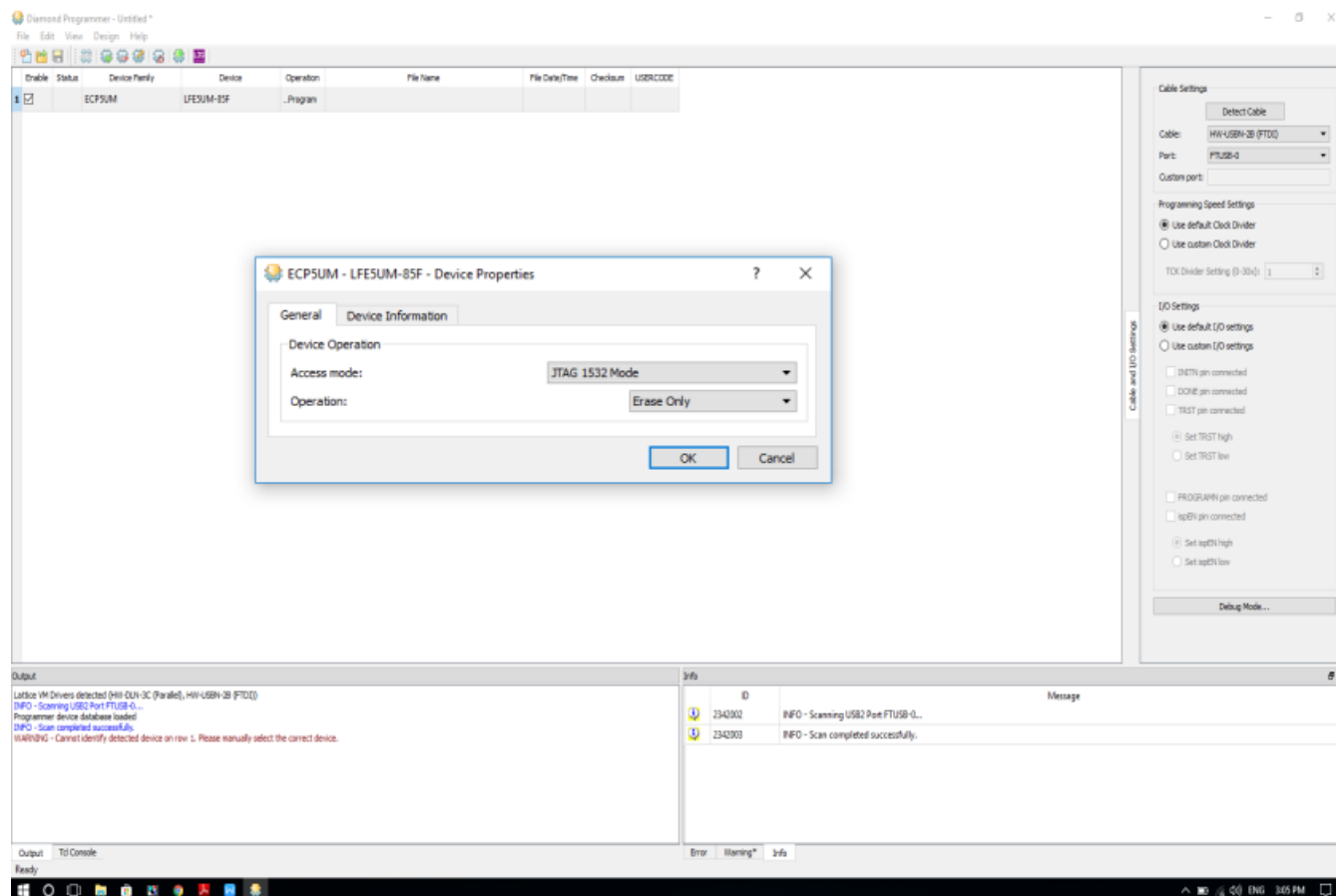
**Figure 23. FTDI USB Cable Detection**

- The FPGA device (LFE5UM-85F) is detected in the programmer window, as shown in [Figure 24](#).



**Figure 24. FPGA Device Scan**

- If the command line prompts to select the device manually, then click on the Device column and select the LFE5UM-85F device.
- Click the Operation tab, and select the Access mode options as JTAG 1532 mode; and for Operation select “Erase Only” for the Erase bit file in the FPGA, as shown in [Figure 25](#).



**Figure 25. Select JTAG Option and Bit File**

5. Click the Program option to complete the process.
6. Click the Operation tab to open the window, as shown in [Figure 26](#).
  - a. Set the Access mode option as SPI Flash Background Programming.
  - b. Set the Operation as “SPI Flash Erase, Program, Verify” for bit file programming.
  - c. Select the bit file from the following path with the mmWave studio package:  
 \mmwave\_studio\_xx\_yy\mmWaveStudio\DCA1000FPGA\DCA1000\_FPGA\_RECORD\_xx\_yy.bit  
 file in the programming file option. (xx -> Version No, yy -> Release Date).
  - d. Select SPI Flash device as (Family: SPI Serial Flash, Vendor: Micron, Device: SPI-N25Q128A, Package: 16-pin SO16)



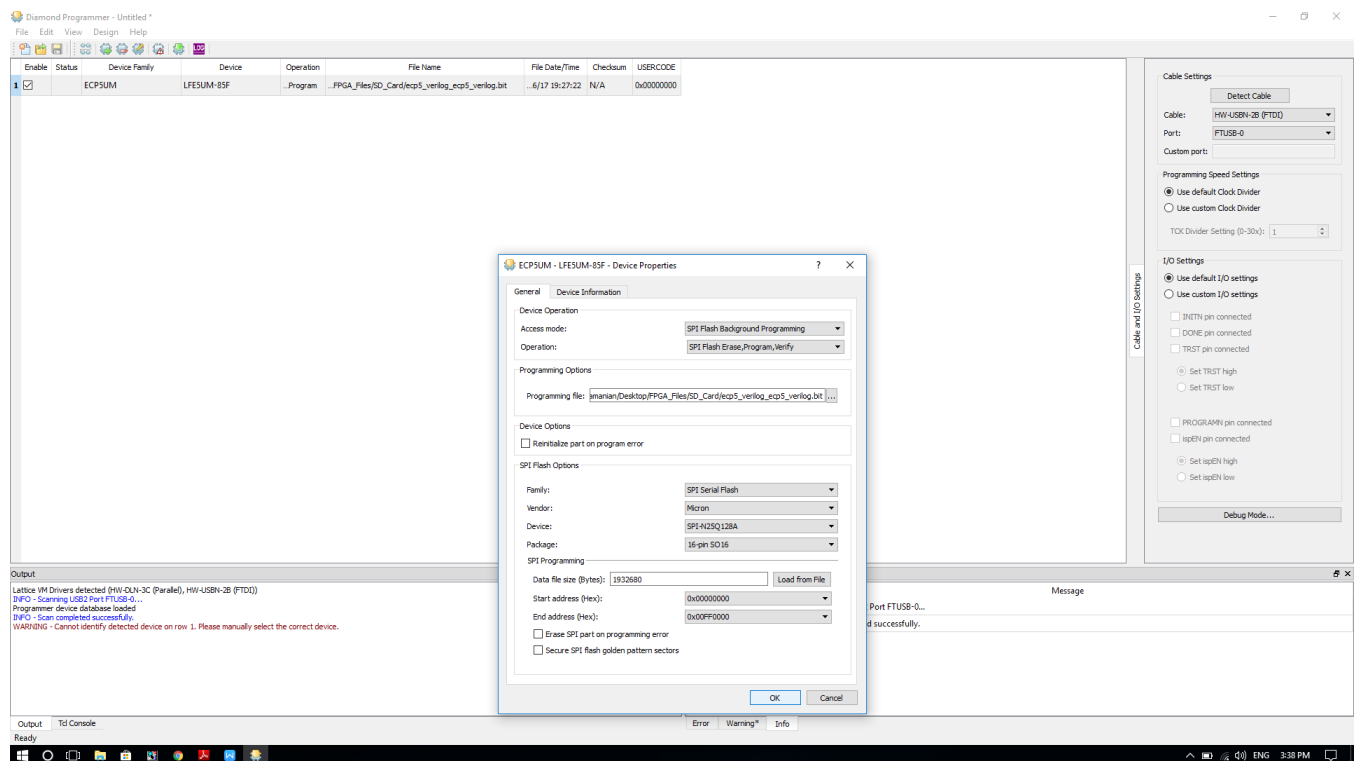


Figure 26. FPGA SPI Flash Programming

7. Click the Program option to complete the process.
8. When the programming is done, press the PROGRAMN(SW4) button or power cycle the board to load the FPGA bit file from SPI flash.
9. The FPGA DONE LED (LD14) glows to confirm the successful loading of the FPGA bit file from SPI flash.

## 10 Limitations

- This release supports LVDS to Ethernet streaming for raw mode and data segregated mode data capture.
- The SD card storage and playback feature is not supported in this release.
- The DCA1000EVM FPGA requires a minimum delay of 12 ms between the bit clock starts and the actual LVDS data start to lock the LVDS PLL IP.
- In default conditions, Ethernet throughput varies up to 325 Mbps speed in a 25- $\mu$ s Ethernet packet delay. The user can change the Ethernet packet delay from 5  $\mu$ s to 500  $\mu$ s to achieve different throughputs, as explained in [Table 19](#), to achieve different Ethernet throughputs.

Table 19. Theoretical Throughput

| Ethernet Packet Delay | Theoretical Throughput (Mbps) |
|-----------------------|-------------------------------|
| 5 $\mu$ s             | ~706                          |
| 10 $\mu$ s            | ~545                          |
| 25 $\mu$ s            | ~325                          |
| 50 $\mu$ s            | ~193                          |

## Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

| Changes from Original (May 2018) to A Revision            | Page |
|-----------------------------------------------------------|------|
| • Updated DCA1000EVM Functional Block Diagram. ....       | 4    |
| • Updated FPGA - SPI Flash Programming Mode section. .... | 25   |

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